Chang Lu

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Education

Yale University New Haven, CT
Ph.D. in Computational Biology & Biomedical Informatics | Advisor: Yuval Kluger *Sep. 2022 – Expected May 2028*
- *Teaching Fellow: Introductory Statistics; Probability & Bayesian Statistics; Intermediate Machine Learning*

M.S. in Computational Biology & Biomedical Informatics *Dec. 2025*

Nanjing University | B.S. in Physics; GPA 90/100, rank 3/98 Nanjing, China | *Sep. 2018 – Jun. 2022*

Research Experience

Data-Efficient Machine Learning & Retrieval *2024 – Present*

OntoEL: Knowledge-Guided Re-Ranking for Neural Retrieval [Paper] [Code] *First Author* · **SIGIR 2026 (Oral)**
- Built an entity-linking re-ranker that infers mention types from context and demotes candidates violating knowledge-base rules
- Fine-tuned SapBERT and MedCPT encoders end-to-end with joint ranking and type losses and hard-negative mining
- Benchmarked on 200K+ mentions: +2.8–4.2 pp top-1 accuracy over base retrievers, 30× faster than a cross-encoder re-ranker
- Analyzed 2,492 cases with the correct entity retrieved but misranked, moving it to rank 1 in 71% vs. 43% for a cross-encoder

Label-Efficient Data Acquisition for Fair Active Learning [Paper] [Code] *First Author* · **KDD 2026**
- Established the first lower bounds on how many labels any method needs to verify that a classifier is fair across groups
- Quantified a 90× difference in label needs between fairness definitions in simulation when only 1% of cases are positive
- Designed a near-optimal stratified sampling scheme that covers underrepresented groups without biasing model training
- Evaluated on 3 datasets of up to 195K records, cutting label use 36–39% vs. a passive baseline at equal accuracy and fairness

Noise-Tolerant Rule Learning with Generalization Guarantees [Paper] [Code] *First Author* · **UAI 2026**

LLM Evaluation & Interpretability *2024 – Present*

Causal Auditing of Intersectional Bias in Clinical LLMs [Paper] [Code] *First Author* · **EMNLP 2026 (Oral)**
- Designed a 2×2 factorial experiment on 2,004 matched counterfactual clinical vignettes varying race and HIV status
- Decomposed Gemma-2-9B with sparse autoencoders; ANOVA showed HIV status amplified race-linked stigma activations
- Verified causality by intervening on a race-encoding activation: 42% of risk ratings flipped, yet reasoning never cited race
- Developed a targeted edit that cut the demographic parity gap 73% with minimal accuracy loss, replicated on Llama-3.1-8B

Theory-Guided Verification of Learned Algorithms in Transformers [Paper] [Code] *First Author* · **KR 2026 (Oral)**
- Translated a theory-predicted 3-step algorithm into a causal model, turning the prediction into a falsifiable hypothesis
- Verified all 3 steps in a trained Transformer by intervention: 96–98% match to predictions; negative controls at chance
- Designed a metric that tracks when the predicted algorithm emerges in training, an abrupt shift that weight norms miss

Work in Submission *2 First-Author Manuscripts · Under Review at **ICLR 2027** & **KDD 2027***
- Built a causal test of chain-of-thought faithfulness: reasoning-distilled LLMs commit up to 2.21× earlier than warranted
- Trained Transformers to count graph cycles exactly, sustaining >98% accuracy on graphs 4× larger than training

Statistical Modeling & Inference *2022 – Present*

MOSAIC: Representation Learning for Multi-Modal Cohorts [Paper] [Code] *First Author* · **RECOMB 2026 (Oral)**
- Developed a spectral method that jointly learns per-patient embeddings of high-dimensional features in a shared space
- Designed a calibrated permutation test for changes in feature relationships, not just feature levels, across conditions
- Validated the test in controlled simulations, outperforming the best baseline with AUPRC 0.98 vs. 0.74
- Built a lasso late-fusion classifier for COVID-19 severity, raising cross-validated AUROC from 0.844 to 0.879 over feature levels

Longitudinal Modeling of High-Dimensional Cohort Data *Co-first Author* · **NeuroHIV 2025 (Oral)**
- Analyzed 5,400+ protein biomarkers in 53 participants, including 15 with irregularly timed repeated measurements
- Combined polynomial trajectory fits, detrended residual tests and mixed-effects regression at subject, group and cohort levels
- Controlled false discoveries with per-test BH correction, identifying 97 proteins significant in at least one test
- Collaborated with biologists to address their research questions, presenting statistical findings and jointly interpreting results

Industry Experience

Baidu, AI and Data Platform | *Product Data Analyst Intern* *Jan. 2021 – Feb. 2021*
- Defined daily-active-user, new-user and funnel metrics with deduplication rules; specified event tracking with developers
- Analyzed 76 days of usage in SQL; recommended prioritizing Windows cloud-task features given its 61% share of downloads

Technical Skills

Machine Learning: Deep learning, Transformers, LLMs, representation learning, retrieval/ranking, active learning, interpretability
Programming: Python (PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, SciPy), R, SQL, Bash/Linux, Git
Statistics: Regression and mixed-effects models, multivariate analysis, experimental design, causal inference, hypothesis testing